Building the Lagos Flood Hazard Map

Step-by-step in QGIS 3.44 — every click, with the traps marked. Attah Gbubemi David.

What you are building

Eight conditioning factors, each rescaled to 0–1, combined by AHP weights into a single Flood Susceptibility Index, then cut into five classes. The result reproduces the published map to **95.92% of cells exactly and 100% within one class**.

Setting	Value	Why it is fixed
Coordinate system	EPSG:32631 (WGS 84 / UTM 31N)	metres in both directions; slope, distance and area are meaningless in degrees
Reference grid	6053 columns x 1225 rows	every layer must match this exactly or the overlay is silently wrong
Pixel size	30 m	the DEM's native resolution after reprojection
Extent (xmin, xmax, ymin, ymax)	467520, 649110, 704640, 741390	read off dem.tif; 181,590 / 30 = 6053 and 36,750 / 30 = 1225

Where the files live

Folder	Contents
...\flood_model_bundle\models\factors\	the eight raw conditioning rasters — **read-only inputs, never write here**
Desktop\aligned\	everything you create
Desktop\lagos_demo.qgz	the QGIS project (save early, save often)

Five rules that prevent every mistake in this document

#	Rule
1	**Save the project after every successful step.** Ctrl+S costs a second. A crash without it costs the session.
2	**Read the Log tab before trusting any Warp result.** The GDAL command line is printed there. If it does not say what you expect, the output is wrong even when it says “completed successfully”.
3	**Never point an output at an input.** Always write to Desktop\aligned\. Overwriting a source raster is unrecoverable if it is not backed up.
4	**Check the value range of every layer you create.** Almost every failure here shows up as a range that is obviously wrong — negatives, infinities, or 3.4e38.
5	**Keep expressions to four layers or fewer.** The Raster Calculator crashes on large multi-layer expressions on a memory-tight machine.

Block 1 — Project setup and loading

Why: UTM 31N gives metres. The project CRS controls what gets measured and what CRS new outputs inherit.

1.1 **Project ■ New**, then **Project ■ Save As** → `Desktop\lagos_demo.qgz`

1.2 Bottom-right, click the **EPSG:4326** button. Filter `32631`, choose **WGS 84 / UTM zone 31N**, click **OK**.

1.3 **Layer ■ Add Layer ■ Add Raster Layer**, click the **...** button, paste the factors path, press Enter.

1.4 Ctrl-click all nine: `dem`, `twi`, `slope`, `flow_accumulation`, `distance_to_rivers`, `lulc_10m`, `curve_number`, `rainfall_95pct`, `rainfall_total_2021`. **Open ■ Add ■ Close**.

> ***Expect:*** *Bottom-right reads EPSG:32631. Nine layers in the panel. Zoom to dem shows Lagos dark along the coast, lighter inland.*

> **TRAP — Load each file once**
> Adding the folder twice puts two copies of every layer in the project. The Raster Calculator then lists each name twice and you cannot tell which is which. If you see duplicates, remove them before going further.

Block 2 — Verify the grids

Why: Raster Calculator works cell by cell. If two layers have different extents or pixel sizes, cell 1 is not the same place on the ground in both, and you get a wrong map with no error message.

2.1 Right-click each layer → **Properties ■ Information**. Read **CRS**, **Width**, **Height**, **Pixel Size**.

Layer	Size	Status
dem, twi, slope, flow_accumulation, rainfall_95pct, rainfall_total_2021	6053 x 1225 @ 30 m	the reference grid
distance_to_rivers	6500 x 2833 @ 30 m	right pixel size, wrong extent
curve_number	6053 x **1224**	one row short
lulc_10m	18159 x 3672 @ **10 m**	three times finer

> ***Expect:*** *Five layers already match. Three do not. Those three are Block 3.*

Block 3 — Align the three odd layers

Why: all eight factors must sit on the identical 6053 x 1225 grid before they can be added together. This is the step with the most traps in the whole method.

Create `Desktop\aligned\` first. Then, for each of the three layers, open **Processing Toolbox** (Ctrl+Alt+T), search `warp`, and double-click **GDAL ■ Raster projections ■ Warp (reproject)**.

Field	Set it to
Input layer	click the **...** → **Select File** → browse to the file in `models\factors\` — never the dropdown
Target CRS	click the ■ → **Project CRS: EPSG:32631**
Resampling method	see the table below — it differs per layer
Output file resolution	`30`
Advanced ■ Georeferenced extent	click the ■ → **Calculate from Layer ■ dem**
Advanced ■ CRS of the target raster extent	**EPSG:32631**
Reprojected	Save to File → `Desktop\aligned\<name>.tif`

Layer	Output name	Resampling	Why that method
distance_to_rivers	dist_rivers_30m.tif	**Bilinear**	distance is continuous; averaging 120 m and 150 m gives 135 m, a real distance
curve_number	curve_number_30m.tif	**Nearest neighbour**	CN is a lookup value; averaging 55 and 98 gives 76.5, which matches no real surface

Layer	Output name	Resampling	Why that method
lulc_10m	lulc_30m.tif	**Mode**	nine 10 m cells collapse into one; mode takes the majority class. Averaging class codes 10 and 50 would invent class 30, which is grassland — pure fiction

Before clicking Run, and again after

Open the **Log** tab and read the `gdalwarp` command. It must contain all four of these:

```
-t_srs EPSG:32631   -tr 30.0 30.0   -te_srs EPSG:32631   -r <near|bilinear|mode>
```

> ***Expect:*** `Creating output file that is 6053P x 1225L`, `Process completed successfully`, *and* ***no red lines****. Ranges: dist_rivers_30m 0–26864, curve_number_30m 55–98, lulc_30m 1–5.*

> **TRAP — Target CRS silently defaults to EPSG:4326**
> The dialog opens with 4326 selected. If you leave it, GDAL is told your UTM metres are degrees and prints a thousand `PROJ: utm: Invalid latitude` warnings, then writes a file of the right shape with a corrupt header. It still says "completed successfully". The legend reading min 0 / max 0 is the giveaway.

> **TRAP — The extent CRS is a second, separate setting**
> Under Advanced, **CRS of the target raster extent** also defaults to 4326. Setting Target CRS alone is not enough — both must read 32631, or the extent numbers are interpreted as degrees.

> **TRAP — Never select the output as the input**
> After a run, QGIS auto-loads the result into the project, so it sits in the input dropdown looking like a normal layer. Choosing it and writing to the same path gives `ERROR 5: Source and destination datasets must be different`. Worse, choosing it and writing to the *original* path overwrites your source raster. Use **Select File** and browse, every time.

Block 4 — Build the three derived layers

Why: runoff does not exist yet, flow accumulation is unusable raw, and land cover is categorical so it cannot be rescaled — it has to be scored.

All of these use **Raster ■ Raster Calculator**. Output each to `Desktop\aligned\`, tick **Add result to project**.

4a Potential maximum retention, S

```
(25400 / "curve_number_30m@1") - 254
```

Output `S.tif`. This is the SCS-CN retention term: S = 25400 / CN – 254, in millimetres.

> ***Expect:*** *range roughly 5 to 208*

4b Runoff depth, Q

```
("rainfall_total_2021@1" > 0.2 * "S@1") * ("rainfall_total_2021@1" - 0.2 * "S@1") ^ 2 /
("rainfall_total_2021@1" + 0.8 * "S@1")
```

Output `runoff.tif`. The SCS-CN runoff equation. The leading condition enforces the initial-abstraction rule: no runoff until rainfall exceeds 0.2S.

> ***Expect:*** *range 0 to about 1882 mm*

4c Log-transformed flow accumulation

```
ln(1 + "flow_accumulation@1")
```

Output `logflow.tif`.

TRAP — Flow accumulation must be log-transformed
Raw values run from 1 to over 1.5 million — seven orders of magnitude. Normalised directly, every cell except the main channels rounds to zero and the factor contributes nothing. The log makes the distribution usable. Be ready to explain this; it is a fair question and the answer is convincing.

Expect: *range 0.693 to 14.41*

4d Land-cover susceptibility score

Processing Toolbox ■ Reclassify by table, raster layer `lulc_30m`.

Min	Max	Value	Class	Reasoning
1	1	0.90	Built-up	impervious; rain runs off almost entirely, and fast
2	2	0.30	Shrub / grass	roots and litter; infiltrates well
3	3	0.10	Water	already water, not susceptible land
4	4	0.70	Cropland	tilled and compacted; little interception
5	5	0.75	Wetland	already saturated; no spare storage

Setting	Value
Range boundaries	**min <= value <= max**
Output NoData value	`-9999`
Use NoData when no range matches value	**ticked**
Output data type	Float32
Output	`Desktop\aligned\lulc_score.tif`

TRAP — Both of those settings matter
With min = max on every row, **min < value <= max** asks for a number both greater than 1 and at most 1 — impossible, so nothing matches. And if **Use NoData when no range matches** is left unticked, the class-0 cells outside Lagos pass through as a real value of 0.0 and enter the overlay as valid low-susceptibility land. Your zone areas would then include the Atlantic.

Expect: *exactly five distinct values: 0.10, 0.30, 0.70, 0.75, 0.90. If the legend still shows 1–5, remove and re-add the layer — QGIS caches the old statistics.*

Block 5 — Rescale the seven measured factors to 0–1

Why: the eight factors are measured in metres, degrees, millimetres and unitless indices. They cannot be added until they share a scale. Three run backwards and must be inverted: low elevation, flat slope and short distance to a river are the *hazardous* ends.

Getting the cut points yourself

Right-click a layer → **Properties ■ Symbology** → under **Min / Max Value Settings** click the radio button for **Cumulative count cut**, leave it at 2.0 – 98.0, click **Apply**, and read the Min and Max boxes at the top.

TRAP — Clicking the radio button is the step people miss
Typing 2.0 and 98.0 into the boxes does nothing unless the **Cumulative count cut** radio is actually selected. If Min / max is still selected you are reading the absolute extremes, not the percentiles.

Why percentiles and not the true minimum and maximum: the DEM's absolute range spans 112.75 m, because 2% of Lagos sits below sea level in dredged creek beds and 2% sits on inland ridges. The 2nd–98th percentile range is 41.78 m. Stretch across the absolute range and the entire populated coastal plain is compressed into a narrow slice of the scale, and elevation stops discriminating between the places that matter.

Layer	Output	2nd pct	98th pct	Direction
dem	elev_n	0.0000	41.7758	**inverted**
twi	twi_n	5.0449	22.6801	direct
slope	slope_n	0.0000	6.0195	**inverted**
dist_rivers_30m	dist_n	0.0000	19028.4473	**inverted**
logflow	flow_n	0.6931	8.5718	direct
runoff	runoff_n	0.0000	1778.4250	direct
rainfall_95pct	rain_n	0.0000	420.4656	direct

The two expressions

Direct — higher value means more hazardous:

```
min(max(("LAYER@1" - LO) / (HI - LO), 0), 1)
```

Inverted — lower value means more hazardous:

```
min(max((HI - "LAYER@1") / (HI - LO), 0), 1)
```

Substitute the layer name and the two numbers from the table. The `min` and `max` buttons are in the Operators panel of the calculator. The nesting clamps everything outside the percentile range to 0 or 1.

Worked example — dist_n (inverted, LO = 0, HI = 19028.4473):

```
min(max((19028.4473 - "dist_rivers_30m@1") / 19028.4473, 0), 1)
```

Expect: *every one of the seven outputs has range **0 to 1**. Check each one before moving on. Inside Lagos dist_n sits between about 0.70 and 1.00, because everywhere in the state is within roughly 5.6 km of a waterway.*

TRAP — A range outside 0–1 means the expression is wrong
During preparation one of these produced 1.4 million infinities from a mistyped division. It poisoned the whole index and was only caught two blocks later. Read the range on every single output. It takes five seconds and saves an hour.

lulc_score is already on the 0–1 scale by construction and skips this block entirely.

Block 6 — Derive the AHP weights

Why: the eight factors do not matter equally. AHP replaces one indefensible decision (pick eight numbers) with 28 defensible ones (compare two factors at a time), then audits whether those 28 contradict each other. Spreadsheet, not QGIS.

A ready-made workbook with live formulas is at `AHP_weights.xlsx`. To build it from scratch:

6.1 Headers in B1:I1 and the same eight names down A2:A9: Elev, TWI, DistRiv, FlowAcc, Runoff, Slope, Rain, LULC.

6.2 Fill B2:I9 with the pairwise matrix (Table 3.4). **Type fractions as** `=1/2`**, not** `1/2` — the second is read as a date, silently, and every number downstream is wrong.

	Elev	TWI	DistRiv	FlowAcc	Runoff	Slope	Rain	LULC
Elev	1	1	1	2	2	2	3	4
TWI	1	1	1	2	2	2	2	4
DistRiv	1	1	1	1	1	2	2	3
FlowAcc	=1/2	=1/2	1	1	1	1	2	2
Runoff	=1/2	=1/2	1	1	1	1	2	2
Slope	=1/2	=1/2	=1/2	1	1	1	1	2
Rain	=1/3	=1/2	=1/2	=1/2	=1/2	1	1	2
LULC	=1/4	=1/4	=1/3	=1/2	=1/2	=1/2	=1/2	1

6.3 `J2 = GEOMEAN(B2:I2)`, filled down to J9. `J10 = SUM(J2:J9)`.

6.4 `K2 = J2/$J$10`, filled down. `K10 = SUM(K2:K9)` must be exactly 1.

6.5 `L2 = B2*$K$2 + C2*$K$3 + ... + I2*$K$9`, filled down. Written out term by term because SUMPRODUCT needs both ranges the same shape and this row is horizontal.

6.6 `M2 = L2/K2`, filled down. Then `λmax = AVERAGE(M2:M9)`, `CI = (λmax - 8)/7`, `CR = CI/1.41`.

Factor	Geometric mean	Derived weight	Applied in model
Elevation	1.7692	0.2019	0.20
TWI	1.6818	0.1919	0.18
Distance to rivers	1.3643	0.1557	0.15
Flow accumulation	1.0000	0.1141	0.12
Runoff	1.0000	0.1141	0.12
Slope	0.8409	0.0959	0.10
Extreme rainfall	0.6722	0.0767	0.08
Land cover	0.4358	0.0497	0.05
TOTAL		**1.0000**	**1.00**

> ***Expect:* λ*max = 8.1139* • *CI = 0.0163* • *CR = 0.0115****, against an acceptance threshold of 0.10. If K10 is not exactly 1, a fraction was entered as a date.***

1.41 is Saaty's Random Index for n = 8 — the average CI of a randomly filled matrix. CR = 0.0115 therefore means the judgements are about **1% as inconsistent as random guessing**.

Block 7 — The overlay, in five small steps

Why split it: the single twelve-layer expression crashes the Raster Calculator (`QgsRasterMatrix::twoArgumentOperation`, access violation) on a memory-tight machine. Each step below touches four layers at most.

Close Chrome, Brave, Edge, Zoom and Docker first. Save after every step.

7.1 First half of the weighted sum

```
0.20 * "elev_n@1" + 0.18 * "twi_n@1" + 0.15 * "dist_n@1" + 0.12 * "flow_n@1"
```

Output `sum1.tif`.

> ***Expect:*** *maximum around 0.65*

7.2 Second half

```
0.12 * "runoff_n@1" + 0.10 * "slope_n@1" + 0.08 * "rain_n@1" + 0.05 * "lulc_score@1"
```

Output `sum2.tif`.

> ***Expect:*** *maximum around 0.35. The two halves sum to a maximum of 1.00, as they must.*

7.3 The validity mask

```
("lulc_score@1" > 0) * ("runoff_n@1" >= 0) * ("twi_n@1" >= 0) * ("slope_n@1" >= 0)
```

Output `mask.tif`. A cell counts only if all four of those layers hold a valid value there — the same rule the model applies.

> ***Expect:*** *exactly 0 or 1. Displayed black and white: white is the study area, black is excluded.*

> **TRAP — Without the mask the nodata sentinels are added, not ignored**
> twi_n, slope_n and runoff_n carry -3.4e38 in their nodata cells, and the calculator sums them rather than masking them. That value appears in the index legend as -340282346638... and makes the whole raster unusable.

7.4 The Flood Susceptibility Index

```
"mask@1" * ("sum1@1" + "sum2@1")
```

Output `FSI.tif`. Multiplying by the mask *after* summing is what removes the nodata.

> ***Expect:*** *0 to 0.9930. The genuine values run 0.1550 to 0.9930; the exact zeros are masked-out cells.*

7.5 Five susceptibility classes

Processing Toolbox ■ Reclassify by table on `FSI`.

Minimum	Maximum	Value	Class
0.10	0.20	1	Very Low
0.20	0.40	2	Low
0.40	0.60	3	Moderate
0.60	0.80	4	High
0.80	1.01	5	Very High

Setting	Value
Range boundaries	**min <= value < max**
Output NoData value	`-9999`
Use NoData when no range matches value	**ticked**
Output data type	**Int16**
Output	`Desktop\aligned\hazard_zones.tif`

TRAP — Three numbers in that table are deliberate
0.10 not 0, so the masked-out zeros fall outside every range and become NoData — the lowest genuine index in Lagos is 0.1550, so nothing real is lost. **1.01** not 1.00, so a cell at exactly 1.0 still lands in zone 5. And here the boundary rule is **min <= value < max**, not the <= used in Block 4: a cell at exactly 0.60 belongs to High, not Moderate.

7.6 Colour it

A style file, `hazard_zones.qml`, sits beside the raster. QGIS applies it automatically on load — no Symbology work needed. If you are styling by hand: Paletted/Unique values, Classify, then 1 `#2c7bb6`, 2 `#abd9e9`, 3 `#ffffbf`, 4 `#fdae61`, 5 `#d7191c`.

Block 8 — Exposure: why this step is not reproducible here

The method is zonal statistics: sum gridded population within each hazard zone. The published result is 4,810,069 people, 38.7% of Lagos, in High or Very High.

TRAP — The bundled WorldPop raster cannot produce that number
`models\factors\worldpop.tif` totals **97,528,513** against Lagos's actual 12,421,056 — inflated **7.84x**. It sits on a 30 m grid, but WorldPop publishes at 100 m, and resampling a *count* raster replicates counts rather than splitting them. It is not fixable by rescaling: the resampling also smeared dense urban population across zone boundaries, so the proportions are wrong too (32.6% versus the true 38.7%). Feeding the *published* hazard map into it reproduces the same wrong answer, which proves the fault is the population file.

The Chapter 4 exposure figures came from the original pipeline using native-resolution WorldPop, which is not in the bundle. To reproduce them you would need a 100 m WorldPop Lagos clip. Until then, quote the validated figures from `exposure_by_zone.csv` and say why.

Block 9 — Validate the map

Why: a map that looks right is not evidence. This scores it against an independent observation the model never saw — the MODIS 250 m flood extent for Lagos, July to November 2012 (NASA/DFO via HDX).

Run `flood_model_bundle\6_validate_my_rebuild.bat`. It runs the thesis Test A twice, on the published map and on yours, and prints them side by side. About 90 seconds.

Measure	Published	Rebuild	Reading
Valid cells	4,202,132	4,202,265	133 cells apart in 4.2 million
AUC-ROC	0.620	0.620	identical to three decimals
Sensitivity	0.685	0.684	—
Enrichment, Low	0.52	0.52	flooding depleted here
Enrichment, Moderate	0.66	0.67	—
Enrichment, High	1.19	1.21	—
Enrichment, Very High	1.61	1.59	1.6x more flooding than chance
Cell-for-cell agreement	—	**95.92%**	—
Within one class	—	**100.00%**	no cell off by more than one class

AUC 0.62 is below the 0.70 target, for both maps equally. The map ranks hazard correctly — enrichment rises monotonically across all five classes — but does not discriminate sharply. MODIS at 250 m cannot see street-scale urban flooding, and a single 2012 event is being tested against a static susceptibility surface. Report it; do not defend it.

Every trap in one list

Where	What goes wrong	The tell
Block 1	layers loaded twice	duplicate names in the calculator
Block 3	Target CRS left at 4326	1000+ *Invalid latitude* warnings; legend 0 to 0
Block 3	extent CRS left at 4326	`-te_srs EPSG:4326` in the log
Block 3	output selected as input	*Source and destination must be different*
Block 3	output pointed at a source file	*Permission denied*, then a corrupted input
Block 4	wrong boundary rule on class codes	output is entirely NoData
Block 4	NoData checkbox unticked	ocean enters the index as valid land
Block 5	percentile radio not selected	reading absolute min/max instead
Block 5	mistyped expression	range outside 0–1, or infinities
Block 6	`1/2` typed without =	Excel shows a date; weights do not sum to 1
Block 7	all twelve layers in one expression	QGIS crashes outright
Block 7	no validity mask	-3.4e38 appears in the index
Block 7	zone 1 starting at 0	masked cells classify as Very Low
Any	not saving the project	a crash costs the whole session